

File 2: Report, as PDF. Contains your name and signature seed at the top, all labeled Part A to C figures with settings captions, your Part D process log (three or more prompt versions), and your 250 to 350 word reflection. Every claim in the reflection must cite a figure by number.

Name: Mary Ann Mastri

Signature seed: 93831

Part A

Prompt Version 1 (Part A baseline)

FIXED_PROMPT = "A green sea turtle resting on the beach at sunset, photorealistic"

This was my starting prompt for all of Part A.

Part A1 – Guidance Sweep

Figure A1-a (Guidance = 3)

Filename: AA_1_guidance_3_93831_30_3.png

Caption: Low guidance image produced is softer and allowed freer interpretation of the prompt.

Figure A1-b (Guidance = 7)

Filename: AA_1_guidance_7_93831_30_7.png

Caption: Medium guidance produced somewhat vivid colors and more contrast, and finer grain sand detail with image similar to Figure A1-a.

Figure A1-c (Guidance = 12)

Filename: outputs/AA_1_guidance_12_93831_30_12.png

Caption: High guidance produced oversaturated colors and extreme unrealistic. contrast

A2 – Steps Sweep

Figure A2-a (Steps = 10)

Filename: A2_steps_10_93831_10_7.png

Caption: With only 10 diffusion steps, the image is blurry, lack proper detail. The sand has some focus but like the turtle shell is mostly blurry.

Figure A2-b (Steps = 25)

Filename: A2_steps_25_93831_25_7.png

Caption: At 25 steps, the image is clearer and turtle detail has improved. Colors are more saturated.

Figures A2-c (Steps = 50)

Filename: A2_steps_50_93831_50_7.png

Caption: At 50 steps, image is refined yet is both slighter sharper. Improvements are subtle. Image is still morphing.

A3

Figure A3-a (Seed 1)

Filename: A3_seed_1_1_25_7.png

Caption: Seed 1 creates a baseline image of turtle, and beach with lighting coming from behind turtle.

Figure A3-b (Seed 22)

Filename: A3_seed_22_22_25_7.png

Caption: Seed 22 generates a markedly different image. The turtles position has changed, the angle of the image increased to show more beach and the setting sun.

FigureA3-c (Seed333)

Filename: A3_seed_333_333_25_7.png

Caption: Seed 333 produced a dramatically different image. Turtle position, lighting, color saturation, camera angle are different.

A4 – Negative Prompt

Figure A4 – before – No Negative Prompt

Filename: A4_before_93831_30_7.png

Caption: Without a negative prompt, the image contains artifacts, softer sand, no ocean.

Figure A4-after – With Negative Prompt

Filename: A4_after_93831_30_7.png

Caption: Adding a negative prompt created a cleaner image. There are still artifacts and the image is still evolving, the edges are sharper, ocean is in the background. The scene is more realistic..

Part B – Edit your own photo – Image-to-Image

Figure B

Filename: B_img2img_str0_6_93831_30_7.png

Caption: img2img result using strength 0.6

Part C – Trigger One Failure Mode

Filename C_counting_93831-30-7.png

Caption: Counting failure: Model does not produce exactly eight turtles and the turtles produced are headless and tailless.

Part D – Process Log and Reflection

Process Log (Three Prompt Versions)

Version 1 – Base Prompt

Prompt: “A green sea turtle resting on the beach at sunset, photorealistic.”

I began with minimal prompt detail to ensure consistency across all sweeps iterations. This base version produced somewhat stable images but did not have sky or sun detail.

Version 2 – Added Environmental Detail.

Prompt: “A green sea turtle resting on the beach at sunset, photorealistic, detailed sand texture, warm lighting.”

After seeing softer textures in low-step and low-guidance output images, I added descriptors so the model could generate clearer sand and more specific lighting. (Figure A1-a)

Version 3 – Added Photography and Realistic Cues

Prompt: “A green sea turtle resting on the beach at sunset, photorealistic, detailed sand texture, warm lighting, high-resolution wildlife photography.

Higher steps produced more detail so I updated the prompt to make the model more realistic. This helped to stabilize the images across seeds. (Figure A2-c)

Reflection

Across Parts A-C, I tested how Stable Diffusion responds to guidance, steps, seeds, negative prompts, img2img, and failure-mode prompting. In the guidance sweep, low guidance produced softer and less defined images. In Figure A1-a, guidance 3 resulted in muted colors and less crisp turtle. Increasing the guidance sharpened the details. Figure A1-b created a balanced, somewhat realistic scene with vivid colors. However, Figures A1-c at guidance 12 heavily oversaturated the turtle distorting what is real.

The steps sweep tested how diffusion iterations influence detail. Figure A2-a (10-steps) created a an image that was a semblance of a turtle shell with weird and blurry saturated colors. Figure A2-b(25 steps) show the image is clearer, colors only slightly saturated and overall more realistic detail. Figure A2-c(50 steps) refined the image and both regressed and progressed – adding a head (without eyes) to the tail end of the turtle!

The seed sweep created images with the most dramatic change. Figure A3-a, A3-b, and A3-c are all completely different image even though they all contain identical settings. This test showed how the seeds control the initial noise, and how it will influence location, lighting, angle, background, etc.

Negative prompting had a cleaner result. Figure A4-before had some minor artifact and softer edges, while Figure A4-after was more refined in detail on the fins – thought it did remove the turtle’s tail.

In Part B, the img2img result (Figure B) kept the general shape of the turtle photo I uploaded but it took liberties in defining the turtle’s anatomy. The 0.6 strength seemed to keep the general shape of the turtle and transform it creatively with a limb amputation, long neck and weird skin.

Finally, triggering one failure test (Figure C) showed that stable diffusion struggles with counting. I prompted for exactly 8 turtles and the model created an incorrect number of headless turtles. This inability to count is a known limitation.

Overall, I initially ran out of GPU time and had to wait a few days to get credits back. Then I had Copilot check my code for typos and correctness before using GPU.